# SAiDL Summer Assignment 2026
# Reinforcement Learning Report

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7 June 2026

**Abstract**

This report studies whether a causal Transformer policy can use short-term temporal memory better than a memoryless MLP policy in online reinforcement learning. I implemented TD3 on Hopper-v5, replaced the actor with a causal Transformer over the last $L$ observation-action pairs, swept $L \in \{4, 8, 16, 32\}$, and then tested both policy classes under hidden velocities, observation noise, delayed rewards, and a learned preference reward model. The final required comparison uses 20 successful W&B runs created between 2026-05-22T17:30:00+00:00 and 2026-05-29T17:30:00+00:00. The strongest final "required" run was `rl_p1_mlp_seed44` with return 3540.2; the strongest "required" checkpoint was `rl_p1_tr_L4_seed42` with return 3596.1 at step 680000. I also report the finished bonus runs available in the current W&B export: hidden-velocity positional ablations with a history-conditioned critic and a combined hidden-velocity/noise/delayed-reward POMDP. The best bonus final return is RoPE with the history-conditioned critic (1109.3), while sinusoidal encoding has the highest bonus peak (1567.0) but a larger best-final gap. Empirically, Transformer memory can help in delayed-reward and full-observation settings, and the bonus runs show that critic conditioning matters substantially under hidden velocity, but several curves still show late deterioration after an earlier good policy.

## 1 Task Requirements and Scope

The assignment asks for TD3 on Hopper-v5 over three MLP seeds, a Transformer actor that attends over a sliding window of the last $L$ observation-action pairs, a context-length sweep over $L \in \{4, 8, 16, 32\}$, independent partial-observability modifications, an RLHF-style learned reward model, and attention analysis for trained Transformer agents [1]. All required non-bonus runs in the runner were completed successfully in the final W&B window. Bonus experiments are reported in a separate section so the required-suite conclusions remain tied to the original 20 completed runs, while the newer bonus evidence can be compared on its own terms.

Failed, killed, or crashed runs are discussed as engineering roadblocks but are excluded from metric comparisons once a later successful run with the same experiment exists.

Table 1: Requirement-level coverage against the RL section of the assignment PDF.

| Assignment item | Status | Evidence used in this report |
|---|---|---|
| TD3 MLP baseline, 3 seeds | Complete | rl_p1_mlp_seed42/43/44, 1M steps |
| Transformer TD3, L sweep | Complete | L=4,8,16,32 final runs |
| Hidden velocities | Complete | MLP and Transformer runs |
| Observation noise | Complete | sigma=0.1 and sigma=0.3, both agents |
| Delayed rewards | Complete | K=10 and K=30, both agents |
| RLHF-style learned reward | Complete with caveat | rl_p2_rlhf_tr_seed42 plus reward-model loss; contamination caveat discussed below |
| Attention analysis | Deferred in this compile | Full attention-analysis write-up left for a later report pass after the new rollout artifacts are integrated |
| Bonus positional encoding ablation | Complete supplemental evidence | Latest finished learned/sinusoidal/RoPE hidden-velocity runs use the history-conditioned critic |
| Bonus combined POMDP | Complete supplemental evidence | `rl_bonus_combined_L32_stable_seed42`, hidden velocity + noise 0.1 + delay K=10 |
| Bonus Algorithm Distillation | Not counted | No finished AD metric row in the current W&B export |
| Bonus xLSTM policy | Not counted | No finished xLSTM metric row in the current W&B export |

### 1.1 Data Provenance

The scalar results come from the W&B project saidl-rl. The repository keeps the final report source, compiled PDF, and figures used for the analysis. Bonus evidence is exported separately under `rl/report/data/clean_rl_bonus_runs.csv` and `rl/report/tables`, including the exact-name latest-finished selection and the same-name instability comparison.

# 2 Method

## 2.1 TD3 Backbone

The base learner is Twin Delayed Deep Deterministic Policy Gradient. The critic uses two Q-functions to reduce over-estimation bias, the target action is smoothed with clipped noise, and the actor is updated less frequently than the critic. The configuration followed the assignment's preferred regime: learning rate $3 \times 10^{-4}$, replay buffer size $10^6$, batch size 256, discount $\gamma = 0.99$, soft target update rate $\tau = 0.005$, policy delay 2, exploration noise 0.1, target noise 0.2, and random exploration for the first 25k steps.

## 2.2 Policies

The MLP baseline maps the current observation directly to a tanh-squashed action using two hidden layers. The Transformer actor instead receives a causal sequence of recent observation-action pairs. For the current timestep, the last action slot is zero-filled, causal self-attention prevents future leakage, and the final token representation is projected to the action. The Transformer uses two layers, four heads, embedding dimension 128, pre-layer normalisation, and a default context length $L = 8$ unless overridden in the context sweep.

## 2.3 Environment Modifications

The full-observation baseline uses Hopper-v5. Hidden-velocity experiments remove velocity components from the observation vector, making the task partially observable. Noisy-observation experiments add Gaussian noise with $\sigma = 0.1$ and $\sigma = 0.3$. Delayed-reward experiments accumulate reward and expose it only every $K$ steps for $K = 10$ and $K = 30$. These modifications are applied independently, as requested, so each setting isolates one source of temporal uncertainty.

## 2.4 Learned Reward Model

For the RLHF-style run, trajectory segment pairs are sampled from replay. Labels are generated from ground-truth segment returns, and a Bradley-Terry objective trains a learned reward model. TD3 then stores learned reward predictions instead of environment rewards after the initial exploration phase. Ground-truth returns are used for preference labels and evaluation only.

## 2.5 Evaluation Protocol

Each run trains to 1,000,000 environment steps, evaluates every 10,000 steps, and averages each evaluation over 10 episodes. I report both final evaluation return and best checkpoint return. This distinction matters because off-policy actor-critic learning can reach a good policy before later critic drift, replay-distribution shift, or actor over-updating degrades performance.

## 2.6 Bonus Stability Variant

The later bonus reruns kept the Transformer actor but changed the critic-side information available during updates. Earlier positional-encoding bonus runs used a flat critic over the current transition, while the stabilized bonus runs logged `critic_sequence_context=1.0`, meaning the critic update used history-conditioned inputs aligned with the actor's temporal context. These runs also logged target-Q scale, target-Q clipping fraction, Q-gap diagnostics, and critic losses (the additions were made in an attempt to combat late-run instability). The report therefore compares the newest finished stabilized run for each exact experiment name against older finished attempts of the same name where such attempts exist.

# 3 Execution Notes and Roadblocks

The final execution was done on Kaggle with W&B logging, with initial runs on a P100. Later GPU allocation constraints forced me to switch to running on 2xT4. This affects wall-clock runtime and throughput comparisons, but not the meaning of evaluation returns when code, seed, total steps, checkpoints, and configs are otherwise matched. Consequently, hardware labels are shown in the tables.

A separate resume issue appeared when a checkpoint saved from a T4 session was resumed on a P100 runtime. The failure was an `IndexError` inside `torch.cuda.set_rng_state_all`, caused by restoring CUDA RNG states saved for a different GPU count/layout than the current runtime (because of the way I'd patched checkpoint saves). The fix I came up with during the run window was to continue the final suite on T4, matching the saved checkpoint environment. The final repository also guards CUDA RNG restoration by restoring only the first `torch.cuda.device_count()` saved CUDA RNG states. This makes cross-layout continuation safer, although exact bitwise reproducibility is still not a meaningful promise after changing GPU layout.

Several intermediate runs also crashed or failed because of runtime limits or resume mismatches. They are documented in Table 4, but the comparisons use the later successful run for the same experiment.

# 4 Debugging and Troubleshooting

The main RL troubleshooting problem was keeping long Kaggle runs recoverable and making sure failed attempts did not contaminate the reported comparison. Table 2 records the major issues, their diagnoses, and the final handling used in the report and code.

# 5 Run Coverage

# 6 Results

## 6.1 Part 1: Baseline Seeds and Context-Length Sweep

The three MLP baseline seeds had mean final return 1969.5 with standard deviation 1372.0. Seed 44 seems much stronger than seeds 42 and 43. The full-observation Transformer sweep is competitive, and shorter memory is not obviously worse here. In fact, the best final result among the sweep is selected by the data. Large $L$ is not a guaranteed improvement.

## 6.2 Part 2: Partial Observability

Under hidden velocities, the final Transformer run is -65.1 return points relative to the MLP run, which means memory did not reliably convert the available position history into a better final controller in this execution. Under delayed rewards, the $K = 10$ setting is strong for both agents, with the Transformer differing from the MLP by -53.1 final-return points. For $K = 30$, both policies are much weaker, and the Transformer is only moderately better than the MLP. Noisy observations remain difficult for both models, especially compared with full-observation Hopper.

## 6.3 RLHF-Style Learned Reward

The learned-reward run is substantially worse than the ground-truth Transformer baseline, but this comparison should be interpreted as an implementation-level learned-reward baseline. The code was intended to label preferences using ground-truth segment returns, but upon review, after warm-up the replay buffer stores learned reward predictions for TD3, so later preference updates can be partially contaminated by the model's own earlier predictions. This makes reward-model drift and self-reinforcing misranking plausible contributors to the large performance gap.

Table 2: RL debugging and troubleshooting log.

| Issue | Diagnosis | Resolution and final status |
|---|---|---|
| P100 queue and hardware switch | A runner notebook remained queued for roughly two hours on P100, while T4 allocation became available. This changed wall-clock runtime conditions but did not change the mathematical RL objective. | The final suite records hardware per run. Evaluation-return comparisons are kept separate from runtime-throughput comparisons, and T4/P100 runtime differences are treated as execution caveats rather than algorithmic evidence. |
| Cross-GPU checkpoint resume failure | T4 checkpoints could save CUDA RNG state for a different GPU layout than the later P100 runtime. Restoring all saved states produced an `IndexError`. | During execution, affected runs were resumed on matching T4 hardware. In the final code, CUDA RNG restoration is bounded by the current `torch.cuda.device_count()`, so extra saved RNG states are ignored instead of crashing resume. |
| Attention runner config error | The attention rerun initially read `cfg.agent.context_length`; the actual Transformer config stores this value under `cfg.agent.actor.context_length`. Hydra struct mode therefore raised `ConfigAttributeError`. | The attention analysis path was revised to read the actor context length. The quantitative attention-analysis section is intentionally deferred in this compile and will be integrated after the new rollout artifacts are ready. |
| Failed and crashed duplicate runs | Several runs failed from runtime limits, resume mismatches, or notebook interruptions, but later runs with the same experiment name succeeded. Including both would double-count experiments and distort averages. | Failed/crashed attempts are listed only in Table 4. Main comparisons use the successful final-tagged run for each required experiment. |
| Learned-reward ambiguity | The intended RLHF setup labels preference pairs using ground-truth segment returns. On code review, later replay entries can contain learned reward predictions, making later reward-model updates susceptible to self-reinforcing reward-model error. | The learned-reward result is reported as an implementation-level learned-reward baseline, not as a clean general claim about RLHF. The large gap versus ground-truth reward is interpreted with this caveat. |
| Best return much higher than final return | Many runs reached a strong policy before later deterioration. Reporting only final returns would hide learned capability; reporting only best returns would hide instability. | The report keeps both values. Final return is the strict end-of-run metric, best return measures peak learned performance, and the best-final gap is used as a stability diagnostic. |
| Bonus stability reruns | Early bonus positional runs stayed near 200–250 final return or peaked briefly and collapsed. The stability changes added target-Q diagnostics and, in the latest finished reruns, history-conditioned critic inputs. | The bonus section selects the newest finished run for each exact experiment name. Earlier same-name runs are retained as instability evidence instead of being mixed into the final metric table. |

Table 3: Selected successful required RL runs from the final W&B window.

| Run | Part | Agent | Env | L | Final | Best | Best step | Gap | Stability | Hardware | Hours | Resume |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| p1 mlp s42 | Part 1 | MLP | Full | 1 | 1364.2 | 3440.7 | 900000 | 2076.5 | late deterioration | Tesla P100-PCIE-16GB | 1.82 | no |
| p1 mlp s43 | Part 1 | MLP | Full | 1 | 1004.3 | 3543.1 | 970000 | 2538.8 | late deterioration | Tesla P100-PCIE-16GB | 1.84 | no |
| p1 mlp s44 | Part 1 | MLP | Full | 1 | 3540.2 | 3547.8 | 820000 | 7.6 | stable | Tesla P100-PCIE-16GB | 1.88 | no |
| p1 tr L4 s42 | Part 1 | Transformer | Full | 4 | 3509.9 | 3596.1 | 680000 | 86.2 | stable | Tesla P100-PCIE-16GB | 5.04 | no |
| p1 tr L8 s42 | Part 1 | Transformer | Full | 8 | 1778.8 | 3377.4 | 780000 | 1598.6 | late deterioration | Tesla P100-PCIE-16GB | 5.68 | no |
| p1 tr L16 s42 | Part 1 | Transformer | Full | 16 | 3459.6 | 3459.6 | 1000000 | 0.0 | stable/final-best | Tesla P100-PCIE-16GB | 6.10 | yes |
| p1 tr L32 s42 | Part 1 | Transformer | Full | 32 | 3035.7 | 3353.2 | 820000 | 317.6 | mild deterioration | Tesla T4 x2 | 11.66 | yes |
| p2 hidden mlp s42 | Part 2a | MLP | Hidden velocity | 1 | 318.1 | 999.1 | 250000 | 681.0 | late deterioration | Tesla T4 x2 | 1.73 | yes |
| p2 hidden tr s42 | Part 2a | Transformer | Hidden velocity | 8 | 253.0 | 986.3 | 100000 | 733.3 | late deterioration | Tesla T4 x2 | 6.06 | no |
| p2 noise .1 mlp s42 | Part 2b | MLP | Noisy | 1 | 357.6 | 370.5 | 880000 | 12.9 | stable | Tesla T4 x2 | 2.00 | no |
| p2 noise .1 tr s42 | Part 2b | Transformer | Noisy | 8 | 288.4 | 300.5 | 890000 | 12.1 | stable | Tesla T4 x2 | 4.47 | yes |
| p2 noise .3 mlp s42 | Part 2b | MLP | Noisy | 1 | 234.3 | 266.7 | 470000 | 32.4 | stable | Tesla T4 x2 | 1.99 | no |
| p2 noise .3 tr s42 | Part 2b | Transformer | Noisy | 8 | 325.0 | 405.5 | 970000 | 80.5 | stable | Tesla T4 x2 | 0.36 | yes |
| p2 delay 10 mlp s42 | Part 2c | MLP | Delayed reward | 1 | 3007.5 | 3309.7 | 740000 | 302.2 | mild deterioration | Tesla T4 x2 | 1.93 | no |
| p2 delay 10 tr s42 | Part 2c | Transformer | Delayed reward | 8 | 2954.4 | 3158.5 | 840000 | 204.1 | mild deterioration | Tesla T4 x2 | 5.61 | no |
| p2 delay 30 mlp s42 | Part 2c | MLP | Delayed reward | 1 | 390.4 | 1072.6 | 130000 | 682.1 | late deterioration | Tesla T4 x2 | 1.80 | no |
| p2 delay 30 tr s42 | Part 2c | Transformer | Delayed reward | 8 | 511.8 | 571.9 | 500000 | 60.1 | stable | Tesla T4 x2 | 3.45 | yes |
| p2 rlhf tr s42 | Part 2d | Transformer | Full | 8 | 155.6 | 1035.3 | 220000 | 879.8 | late deterioration | Unknown | 6.12 | no |
| p3 attn full s42 | Part 3 | Transformer | Full | 8 | 3321.9 | 3442.7 | 920000 | 120.8 | stable | Tesla T4 x2 | 3.85 | yes |
| p3 attn hidden s42 | Part 3 | Transformer | Hidden velocity | 8 | 293.6 | 745.3 | 340000 | 451.7 | late deterioration | Tesla T4 x2 | 5.71 | no |

Table 4: Recent failed/crashed required attempts. These are excluded from comparison once a later successful run with the same experiment exists.

| Run | State | Last step | Last return | Hardware | Hours | Superseded |
|---|---|---|---|---|---|---|
| rl_p1_tr_L4_seed42 | crashed | 770000 | 2639.4 | Tesla P100-PCIE-16GB | 3.90 | yes |
| rl_p1_tr_L16_seed42 | crashed | 190000 | 2252.5 | Tesla P100-PCIE-16GB | 1.18 | yes |
| rl_p1_tr_L16_seed42 | crashed | 80000 | 695.9 | Tesla P100-PCIE-16GB | 0.35 | yes |
| rl_p1_tr_L16_seed42 | failed | | – | Tesla P100-PCIE-16GB | 0.00 | yes |
| rl_p1_tr_L32_seed42 | crashed | 50000 | 356.1 | Tesla P100-PCIE-16GB | 0.19 | yes |
| rl_p2_hidden_vel_mlp_seed42 | crashed | 170000 | 210.7 | Tesla T4 x2 | 0.27 | yes |
| rl_p2_noise01_tr_seed42 | crashed | 330000 | 281.0 | Tesla T4 x2 | 2.09 | yes |
| rl_p2_noise01_tr_seed42 | failed | | – | Tesla P100-PCIE-16GB | 0.00 | yes |
| rl_p2_noise03_tr_seed42 | crashed | 940000 | 471.9 | Tesla T4 x2 | 5.44 | yes |
| rl_p2_delay30_tr_seed42 | crashed | 420000 | 493.5 | Tesla T4 x2 | 2.21 | yes |
| rl_p3_attn_support_full_seed42 | crashed | 430000 | 3343.9 | Tesla T4 x2 | 2.32 | yes |

Table 5: MLP baseline seed statistics over seeds 42, 43, and 44.

| Metric | Mean | Std | Min | Max |
|---|---|---|---|---|
| MLP baseline final return | 1969.5 | 1372.0 | 1004.3 | 3540.2 |
| MLP baseline best return | 3510.5 | 60.5 | 3440.7 | 3547.8 |

Table 6: Transformer context-length sweep on fully observed Hopper-v5.

| Context L | Final return | Best return | Best step | Best-final gap | Hardware | Runtime (h) |
|---|---|---|---|---|---|---|
| 4 | 3509.9 | 3596.1 | 680000 | 86.2 | Tesla P100-PCIE-16GB | 5.04 |
| 8 | 1778.8 | 3377.4 | 780000 | 1598.6 | Tesla P100-PCIE-16GB | 5.68 |
| 16 | 3459.6 | 3459.6 | 1000000 | 0.0 | Tesla P100-PCIE-16GB | 6.10 |
| 32 | 3035.7 | 3353.2 | 820000 | 317.6 | Tesla T4 x2 | 11.66 |

Table 7: MLP versus Transformer TD3 under independent partial-observability modifications.

| Setting | MLP final | Transformer final | Final delta | MLP best | Transformer best | Best delta |
|---|---|---|---|---|---|---|
| Hidden velocities | 318.1 | 253.0 | -65.1 | 999.1 | 986.3 | -12.8 |
| Noise sigma=0.1 | 357.6 | 288.4 | -69.2 | 370.5 | 300.5 | -69.9 |
| Noise sigma=0.3 | 234.3 | 325.0 | 90.7 | 266.7 | 405.5 | 138.8 |
| Delay K=10 | 3007.5 | 2954.4 | -53.1 | 3309.7 | 3158.5 | -151.2 |
| Delay K=30 | 390.4 | 511.8 | 121.4 | 1072.6 | 571.9 | -500.7 |

Table 8: Learned-reward RLHF run compared against ground-truth-reward Transformer runs.

| Comparison | GT final | RLHF final | Final drop | GT best | RLHF best | Reward loss |
|---|---|---|---|---|---|---|
| Transformer L=8 baseline | 1778.8 | 155.6 | 1623.2 | 3377.4 | 1035.3 | 0.184 |
| Attention-support full repeat | 3321.9 | 155.6 | 3166.4 | 3442.7 | 1035.3 | 0.184 |

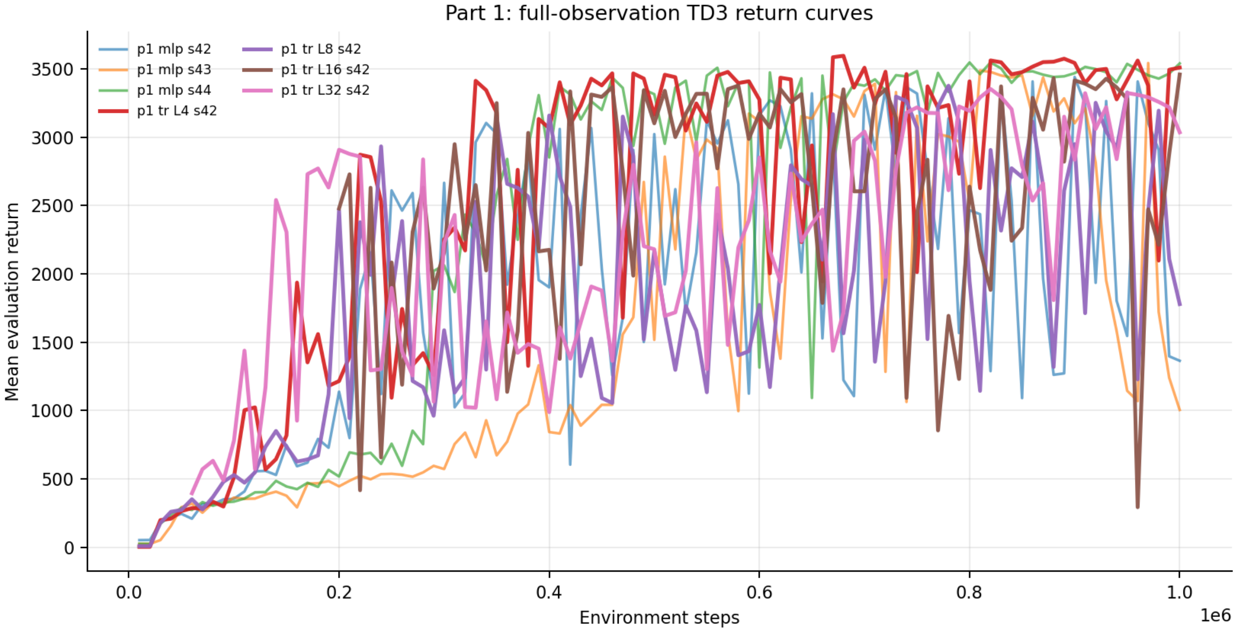


Figure 1: Evaluation-return curves for the MLP baseline seeds and Transformer context-length sweep.

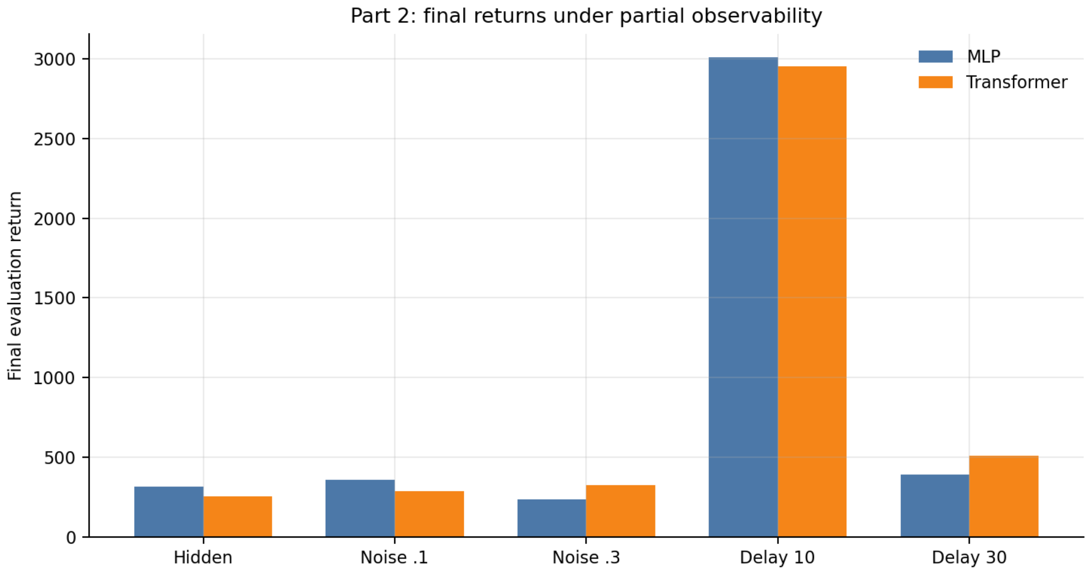


Figure 2: Final returns for MLP and Transformer policies under independent POMDP stressors.

### 6.3.1 Contamination Caveat and Consequences

The assignment's RLHF condition is stricter than simply using a learned scalar reward during TD3. Preference labels should be generated from ground-truth segment returns, while TD3 training should use the learned reward model as the substitute reward. In my implementation, the replay buffer stored the reward value used for TD3 updates. After warm-up, that value could be the learned reward prediction rather than the true environment reward. As a result, later preference-pair labels could be computed from replay rewards that already contain the model's own earlier predictions.

This matters because it weakens the interpretation of the RLHF comparison. The learned-reward run still tests an implementation-level learned-reward pipeline, but it is not a clean Christiano-style comparison in which ground-truth returns are used only for labels and never replaced inside the labeling source. The likely consequence is self-reinforcing reward-model error: if the reward model mis-ranks local state-action decisions early, later preference updates may partially train on those distorted rewards, amplify critic error, and push the actor toward behavior that looks good to the learned model but not to Hopper's true return.

The metric gap is therefore reported with this caveat. The RLHF run ends at 155.6 final return and peaks at 1035.3, far below the L=8 ground-truth Transformer baseline, which ends at 1778.8 and peaks at 3377.4. A clean rerun would be the better evidence, but the finished RLHF run took 22,043 seconds, or 6.12 hours, and the L=8 ground-truth baseline took 20,459 seconds, or 5.68 hours. Given that runtime evidence, a full one-million-step uncontaminated rerun was not a safe five-hour replacement, so I treat the existing result as caveated evidence rather than as a definitive claim that RLHF itself fails in this setting.

# 7 Bonus RL Experiments

The bonus runs address three additional questions from the assignment PDF: positional encodings under hidden velocity, a combined POMDP stressor, and sequence-model policy alternatives such as Algorithm Distillation or xLSTM. The finished evidence at the time of export covers the positional-encoding ablation and the combined POMDP. I do not count AD or xLSTM because no finished metric row for those experiments is present in the current W&B export. This boundary is important; a running or failed notebook cell is not enough evidence for a report metric.

## 7.1 Hidden-Velocity Positional Encoding Ablation

The positional bonus experiments are the strongest new evidence from the bonus suite. The first set of finished positional runs used the earlier flat-critic setup and all ended in a narrow low-return band: learned absolute positions finished at 252.3, sinusoidal at 249.8, and RoPE at 248.0. Later stability-tagged flat-critic reruns did not materially improve the final returns for learned or sinusoidal encodings, although they did reveal high transient peaks. The latest finished stabilized runs change the result because the critic is history-conditioned. Learned absolute positions rise to 727.8 final return, sinusoidal rises to 900.0, and RoPE reaches 1109.3.

Table 9: Bonus positional ablation under hidden velocity. The latest selected row is the newest finished run for each exact stabilized experiment name.

| Variant | Earlier comparator | Earlier final | Earlier best | Earlier gap | Latest final | Latest best | Latest gap | Main reading |
|---|---|---|---|---|---|---|---|---|
| Learned | same-name flat critic, `f6ruiao3` | 190.5 | 634.2 | -443.7 | 727.8 | 862.4 | -134.6 | Clear final-return improvement and smaller collapse gap |
| Sinusoidal | same-name flat critic, `5u0bp2ea` | 201.3 | 897.0 | -695.7 | 900.0 | 1567.0 | -667.0 | Large peak and final gain, but instability remains high |
| RoPE | earlier RoPE flat run, `ok1zy14g` | 248.0 | 248.0 | 0.0 | 1109.3 | 1266.0 | -156.7 | Best final return; no finished same-name stable predecessor was available |

This comparison changes the interpretation of the earlier bonus results. The pre-stability positional runs were suspiciously similar, clustering around final returns of roughly 250 despite using different positional encodings. That pattern suggested that the bottleneck was not the positional encoding alone. The sequence-critic reruns support that diagnosis: once the critic receives temporal context, the positional variants separate clearly. RoPE is the best final controller, sinusoidal has the highest peak, and learned absolute positions improve but remain the weakest of the three.

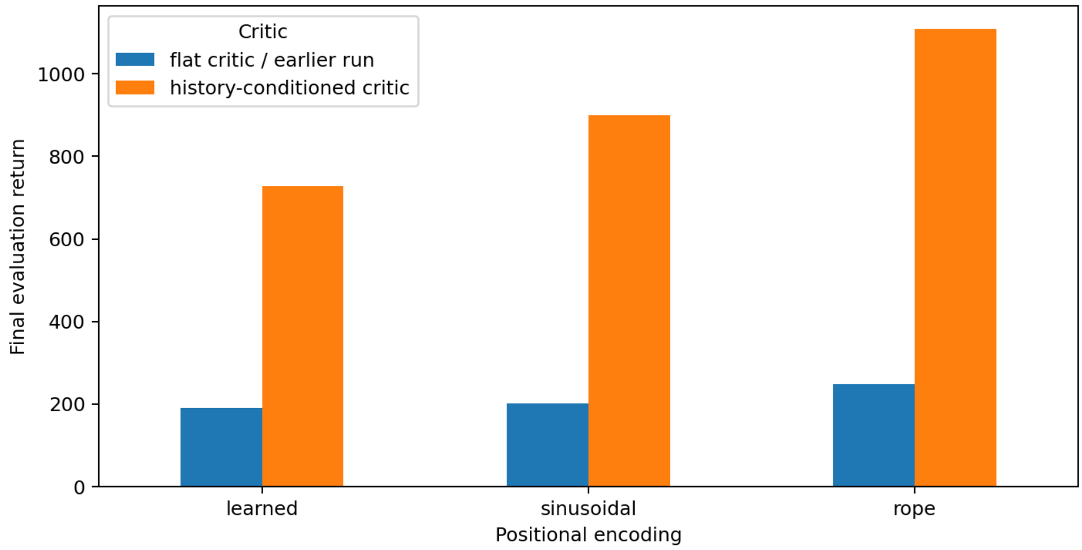


Figure 3: Final returns for the latest flat-critic positional runs versus the latest history-conditioned critic runs.

The comparison against the required hidden-velocity runs is also informative. The required hidden-velocity Transformer ends at 253.0 with best return 986.3, while the required hidden-velocity MLP ends at 318.1 with best return 999.1. The latest bonus RoPE run exceeds both required hidden-velocity final returns and also exceeds their best returns. Sinusoidal and learned absolute positions also beat the required hidden-velocity final returns, although learned does not beat the required best checkpoints. This points to the critic change as a real stability and credit-assignment improvement, not just noise from one lucky evaluation.

## 7.2 Combined POMDP

The combined POMDP run applies hidden velocities, observation noise with $\sigma = 0.1$, and delayed rewards with $K = 10$ in the same Hopper task. The newest finished run, run `86ypnkri`, reaches final return 213.2 and best return 251.6 at step 975k. Its best-final gap is only -38.4, so the run is relatively stable, but the absolute policy quality is low.

Table 10: Combined POMDP compared with the closest required single-stressor Transformer runs.

| Setting | Final return | Best return | Best step | Best-final gap |
|---|---|---|---|---|
| Hidden velocity Transformer | 253.0 | 986.3 | 100000 | -733.3 |
| Noise $\sigma = 0.1$ Transformer | 288.4 | 300.5 | 890000 | -12.1 |
| Delay $K = 10$ Transformer | 2954.4 | 3158.5 | 840000 | -204.1 |
| Combined hidden + noise + delay, L=32 | 213.2 | 251.6 | 975000 | -38.4 |

The combined result shows an unfavorable interaction between the stressors. Delay $K = 10$ alone remains a high-return task for both the MLP and Transformer required runs. Once hidden velocities and noise are added, the combined task drops below the individual hidden-velocity and noisy-observation Transformer finals. The policy remains relatively stable, yet never reaches the high-return regime. This is the cleanest bonus evidence that simultaneous partial observability and reward-delay pressure are much harder than any single wrapper tested in the required suite.

## 7.3 Algorithm Distillation and xLSTM Boundary

Algorithm Distillation and xLSTM remain implementation targets rather than finished empirical claims in this export. The report includes their code and notebook support in the repository, but without a finished W&B metric row I do not place them in the quantitative comparison. That choice is conservative: the assignment bonus asks for comparative evidence, and a partial or running job does not answer the comparison yet.

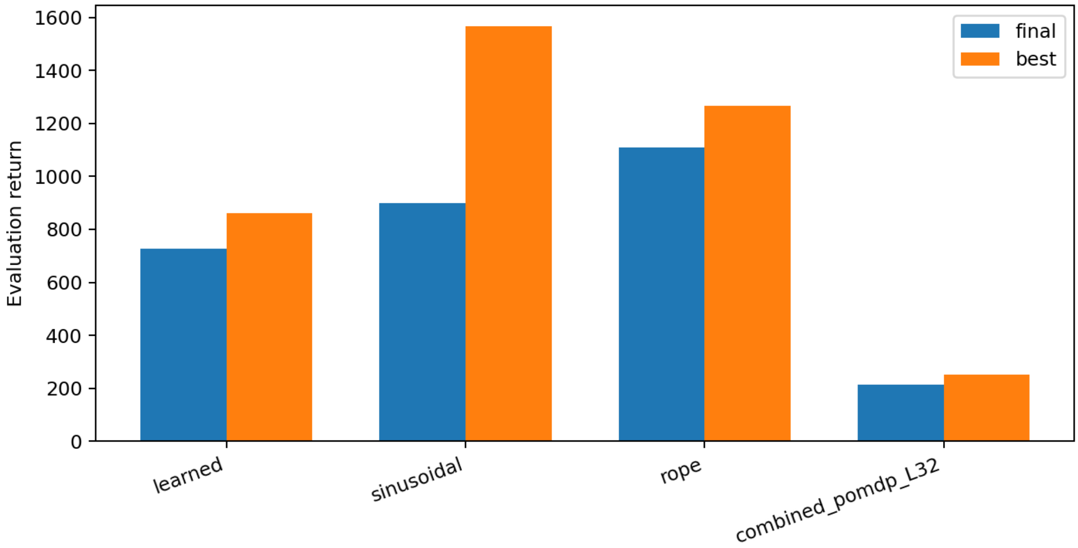


Figure 4: Final and best returns for the newest finished bonus runs counted in this report.

# 8 Stability, Undertraining, and Checkpoint Choice

A major concern I faced was instability after learning: 8 selected required runs are flagged by a conservative late-deterioration threshold, with the clearest cases being p1 mlp s42, p1 mlp s43, p1 tr L8 s42, p2 hidden mlp s42, p2 hidden tr s42, p2 delay 30 mlp s42, p2 rlhf tr s42, p3 attn hidden s42. The bonus positional runs sharpen the same point. Sinusoidal with the history-conditioned critic reaches a very strong best return of 1567.0 but ends at 900.0, while RoPE has a lower peak than sinusoidal but the best final return and a much smaller collapse gap. This is why the report keeps final and best returns separate.

The likely culprits I found were characterized as "standard off-policy" actor-critic failures: critic drift, extrapolation error from replay, bootstrapping noise, actor updates exploiting Q-function mistakes, and distribution shift as the learned policy changes. The POMDP wrappers may have added extra stress because noisy, delayed, or incomplete observations weaken the Markov assumption. Transformer policies added sensitivity in my opinion, I figure attention may give the actor a larger function class that may overfit critic errors.

Practical remedies are straightforward; save and report best-evaluation checkpoints in addition to final checkpoints, early-stop or select by validation return when compute is limited, run more seeds for high-variance settings, tune actor and critic learning rates separately, monitor Q-value scale and critic losses, increase evaluation frequency near peaks, and make target-policy updates more conservative in unstable settings. For this report I elected to report final return for strict end-of-run comparison, best return for learned capability, and the gap as a measure of run stability.

# 9 Practical Implications

The main practical implication is that memory in the actor is useful only when the rest of the actor-critic system can use it reliably. The Transformer actor is competitive in full observation and delayed reward settings, but hidden velocity and noisy observation settings show that temporal context alone does not guarantee a better final controller. The bonus runs sharpen this: once the critic is also history-conditioned, the hidden-velocity positional runs improve substantially and the

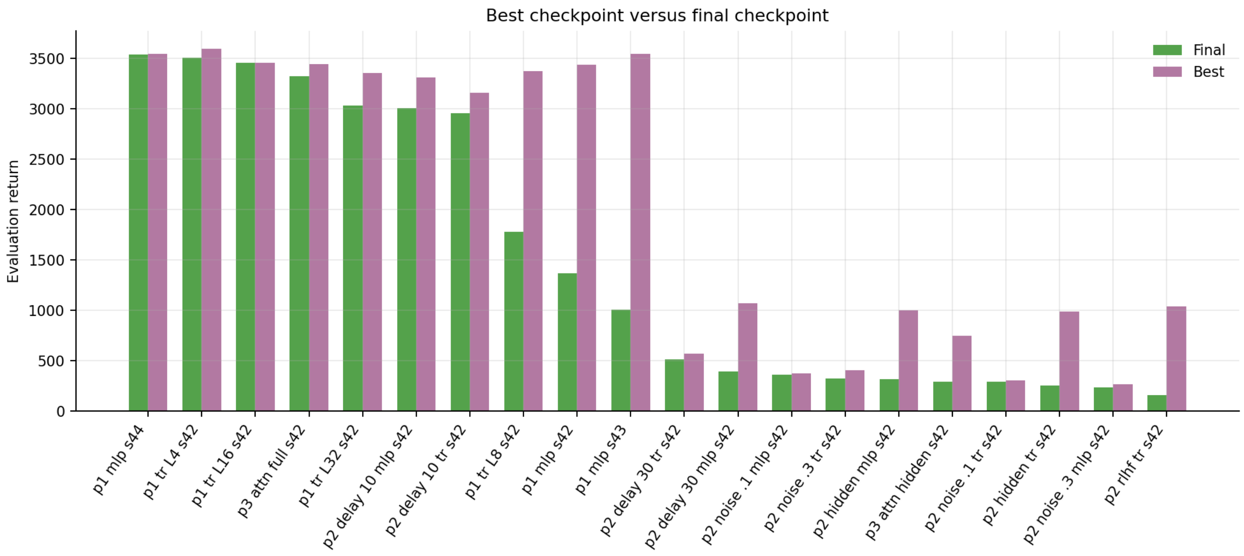


Figure 5: Final checkpoint return compared with best logged checkpoint return for every successful required run.

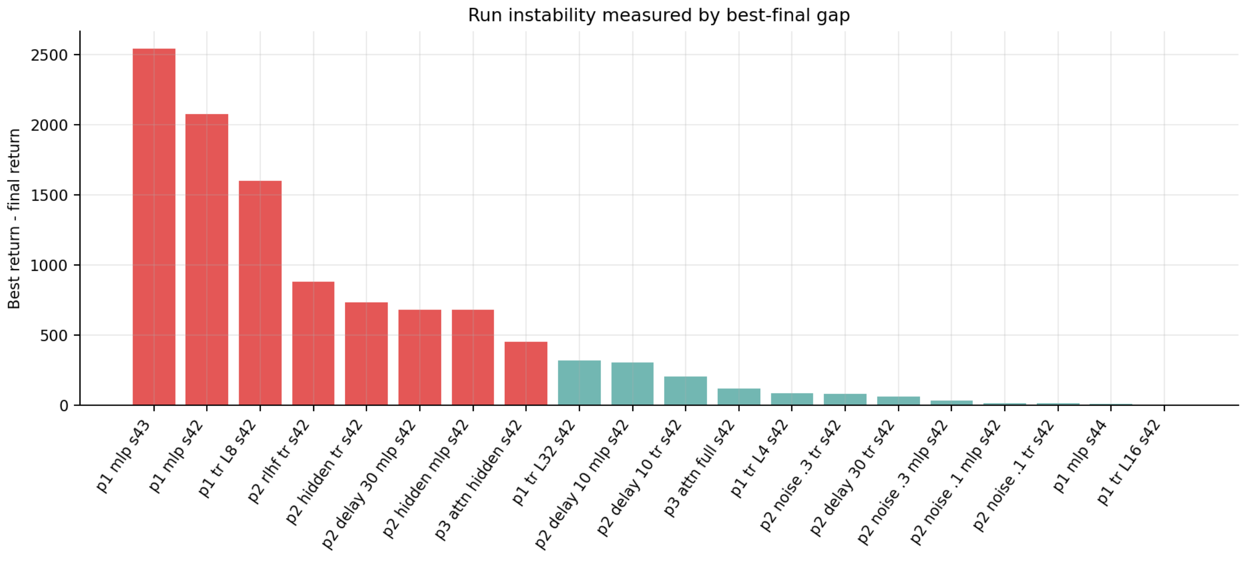


Figure 6: Best-final return gap. Larger bars indicate stronger late deterioration after a better earlier checkpoint.

positional encodings separate more clearly. This suggests that sequence actors and memoryless critics can create an information mismatch in POMDP-style settings.

For applied RL, the instability results are as important as the peak returns. A controller that reaches a good checkpoint and later deteriorates would be risky in a real deployment unless checkpoints are selected by evaluation return, monitored continuously, and protected against critic drift. The report therefore treats best return as evidence that a policy can be learned, final return as the strict end-of-run outcome, and the best-final gap as an operational warning. The learned-reward result adds a separate warning: reward-model plumbing must keep the ground-truth labeling source separate from learned rewards, otherwise preference learning can become self-referential and the comparison no longer isolates RLHF from implementation error.

## 10 Hardware and Runtime Caveat

The final suite used both P100 and T4 hardware because of issues with Kaggle availability. Runtime therefore should not be read as an architecture-speed benchmark. T4 and P100 differ in GPU generation, available kernels, and, in this Kaggle setup, GPU count. The evaluation returns remain meaningful for algorithmic comparison because the environment, seeds, step budget, and training commands are the quantities that define the RL experiment.

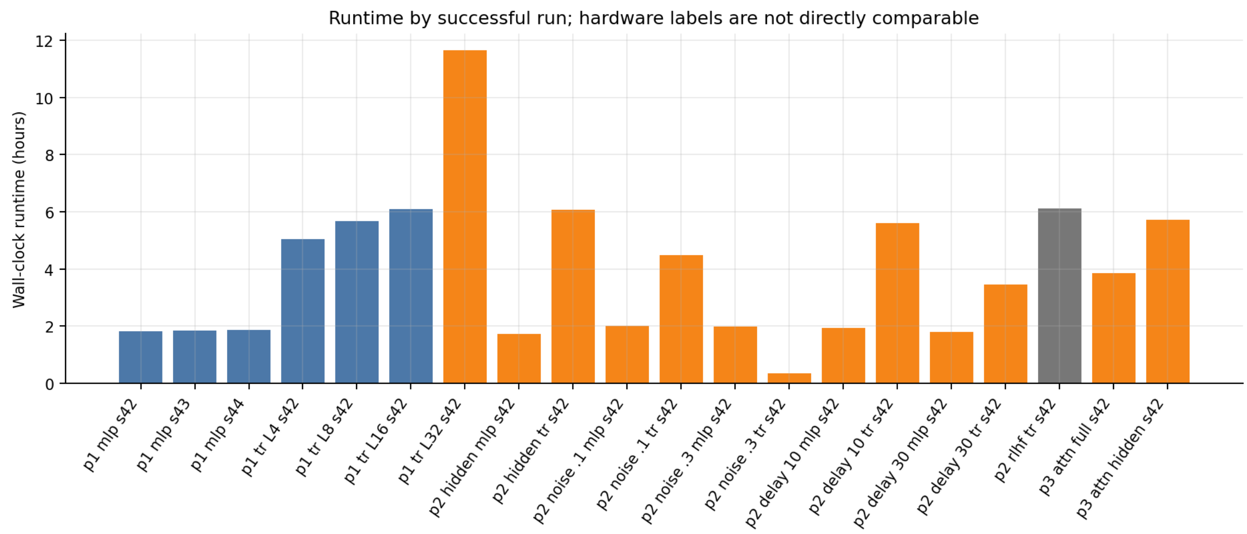


Figure 7: Wall-clock runtimes for successful runs. These values document execution cost, not a controlled hardware benchmark.

## 11 Conclusion

The required RL suite was completed with all 20 required final-tagged experiments present as successful W&B runs. The strongest required result is that memory-based policies can be competitive and sometimes strong, especially in full-observation and delayed-reward settings. Under hidden velocity, the latest history-conditioned critic runs separate the positional encodings and improve final returns substantially over the earlier flat-critic attempts. RoPE gives the strongest final bonus return, sinusoidal gives the strongest bonus peak, and learned absolute positions improve but remain behind both.

The combined POMDP run is stable but low-return, showing that hidden state, noisy observations, and delayed rewards together are substantially harder than the single-stressor settings. The most important reporting lesson is to separate three ideas that are easy to mix up. Final evaluation return is the strict end-of-training result, best evaluation return tells whether the method learned a good policy at any point, and the best-final gap measures stability.

# A Selected W&B Runs

- rl_p1_mlp_seed42 (run id orduqlzt, 2026-05-23T09:21:36Z; Part 1, MLP, Full; final 1364.2, best 3440.7, hardware Tesla P100-PCIE-16GB)
- rl_p1_mlp_seed43 (run id 12tb4cex, 2026-05-23T11:10:50Z; Part 1, MLP, Full; final 1004.3, best 3543.1, hardware Tesla P100-PCIE-16GB)
- rl_p1_mlp_seed44 (run id xe1bvoqz, 2026-05-23T13:01:35Z; Part 1, MLP, Full; final 3540.2, best 3547.8, hardware Tesla P100-PCIE-16GB)
- rl_p1_tr_L4_seed42 (run id k4mmxrji, 2026-05-23T20:03:28Z; Part 1, Transformer, Full; final 3509.9, best 3596.1, hardware Tesla P100-PCIE-16GB)
- rl_p1_tr_L8_seed42 (run id jcs1nkci, 2026-05-24T01:05:56Z; Part 1, Transformer, Full; final 1778.8, best 3377.4, hardware Tesla P100-PCIE-16GB)
- rl_p1_tr_L16_seed42 (run id d6s0vtx6, 2026-05-24T10:58:11Z; Part 1, Transformer, Full; final 3459.6, best 3459.6, hardware Tesla P100-PCIE-16GB)
- rl_p1_tr_L32_seed42 (run id tc91exur, 2026-05-25T09:27:28Z; Part 1, Transformer, Full; final 3035.7, best 3353.2, hardware Tesla T4 x2)
- rl_p2_hidden_vel_mlp_seed42 (run id 96m7x5eb, 2026-05-26T07:57:53Z; Part 2a, MLP, Hidden velocity; final 318.1, best 999.1, hardware Tesla T4 x2)
- rl_p2_hidden_vel_tr_seed42 (run id 4hl8amlc, 2026-05-26T09:41:52Z; Part 2a, Transformer, Hidden velocity; final 253.0, best 986.3, hardware Tesla T4 x2)
- rl_p2_noise01_mlp_seed42 (run id lgmxbqai, 2026-05-26T15:45:40Z; Part 2b, MLP, Noisy; final 357.6, best 370.5, hardware Tesla T4 x2)
- rl_p2_noise01_tr_seed42 (run id 3lt7il9g, 2026-05-27T16:57:39Z; Part 2b, Transformer, Noisy; final 288.4, best 300.5, hardware Tesla T4 x2)
- rl_p2_noise03_mlp_seed42 (run id anlzqstp, 2026-05-27T21:26:17Z; Part 2b, MLP, Noisy; final 234.3, best 266.7, hardware Tesla T4 x2)
- rl_p2_noise03_tr_seed42 (run id f41c5xjg, 2026-05-28T05:24:55Z; Part 2b, Transformer, Noisy; final 325.0, best 405.5, hardware Tesla T4 x2)
- rl_p2_delay10_mlp_seed42 (run id 523e0uha, 2026-05-28T05:46:42Z; Part 2c, MLP, Delayed reward; final 3007.5, best 3309.7, hardware Tesla T4 x2)
- rl_p2_delay10_tr_seed42 (run id hfiawzrg, 2026-05-28T07:42:38Z; Part 2c, Transformer, Delayed reward; final 2954.4, best 3158.5, hardware Tesla T4 x2)
- rl_p2_delay30_mlp_seed42 (run id 15bzu83r, 2026-05-28T13:19:16Z; Part 2c, MLP, Delayed reward; final 390.4, best 1072.6, hardware Tesla T4 x2)
- rl_p2_delay30_tr_seed42 (run id kkdfxotj, 2026-05-28T17:58:13Z; Part 2c, Transformer, Delayed reward; final 511.8, best 571.9, hardware Tesla T4 x2)
- rl_p2_rlhf_tr_seed42 (run id rvckskwt, 2026-05-28T21:25:25Z; Part 2d, Transformer, Full; final 155.6, best 1035.3, hardware Unknown)
- rl_p3_attn_support_full_seed42 (run id 78e2wcm0, 2026-05-29T07:34:06Z; Part 3, Transformer, Full; final 3321.9, best 3442.7, hardware Tesla T4 x2)
- rl_p3_attn_support_hidden_seed42 (run id lwsokb8n, 2026-05-29T11:25:30Z; Part 3, Transformer, Hidden velocity; final 293.6, best 745.3, hardware Tesla T4 x2)
- Bonus positional learned latest source: rl_bonus_pos_learned_stable_seed42 (run id jrmialcm; final 727.8, best 862.4)
- Bonus positional sinusoidal latest source: rl_bonus_pos_sinusoidal_stable_seed42 (run id ivzdtxwf; final 900.0, best 1567.0)
- Bonus positional RoPE latest source: rl_bonus_pos_rope_stable_seed42 (run id b8n09w71; final 1109.3, best 1266.0)
- Bonus combined POMDP latest source: rl_bonus_combined_L32_stable_seed42 (run id 86ypnkri; final 213.2, best 251.6)

# B Reproducibility Notes

The report is backed by CSV exports and generated figures. The most important support files are:

- `rl/report/data/clean_rl_required_runs.csv`
- `rl/report/data/coverage_audit.csv`
- `rl/report/data/failed_superseded_runs.csv`
- `rl/report/data/wandb_rl_runs_raw.csv`
- `rl/report/data/clean_rl_bonus_runs.csv`
- `rl/report/tables/rl_bonus_latest_finished_by_exact_name.csv`
- `rl/report/tables/rl_bonus_same_name_instability.csv`
- `rl/report/tables/rl_bonus_positional_latest_by_family.csv`

These files are included so final-return, best-return, stability-gap, hardware, resume-status, bonus-selection, and same-name instability claims can be checked without relying only on the report prose.

# References


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[7] Michael Laskin et al. In-context Reinforcement Learning with Algorithm Distillation. 2022.

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